Historical Britannica: extracted source

Encyclopaedia Britannica - Third edition, Volume 1, A-ANG - EB.5

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Region region-0; provider order 0.

A L G E B R A.

Region region-1; provider order 1.

Part III.

Region region-2; provider order 2.

Application to Gen metry, -006t nearly, and x=2.094 nearly.

Region region-3; provider order 3.

4. This operation may be continued to any length, as by fuppeling gz=-054+/s, and fo ons, and the value of z=0.094747 nearly.

Region region-4; provider order 4.

By the full operation a never value of x may be found thus i force f=1 nearly and -1+1o/+of/+ i f =0, f=0+of/r/s, that is, f=10+.6+.094 true to the left figure, and x=2.094.

Region region-5; provider order 5.

In the fame manner may the root of a pure equal Application tion be found, and this gives an cafy method of ap- proximating to the roots of numbers which are not perfect powers.

Region region-6; provider order 6.

This can is applicable to numeral equations of every order; and, by alluming a general equation, geneth rules may be deduced for approximating to the roots of any propried equation. By a finalime method we may approximate to the roots of literal equations, which will be exprefiled by infinite ferries.

Region region-7; provider order 7.

P AR T III.

Region region-8; provider order 8.

Of the Application of AlGEBRA TO GEOMETRY.

Region region-9; provider order 9.

CHAP. I.

Region region-10; provider order 10.

General Principles.

Region region-11; provider order 11.

C EOMETRY treats both of the magnitude and polition of extention, and their connections.

Region region-12; provider order 12.

Algebra: tests only of magnitudely, therefore, of the relations which fulfill in geometrical figures, those of magnitude only can be immediately exprefed by algebra.

Region region-13; provider order 13.

The opposite polition of friginal lines may indeed be expelled firmly by the two + and — But, in order to expect the various other positions of geomne- tricical figures by algebra from the principles of geometry, forme relations of magnitude must be found, which depend upon their path, positions, and which can be scalhi- bited by equations ; And, convertily, by the fame principles may the politions and figures is inferred from the equations denoting fish relations of their parts.

Region region-14; provider order 14.

Though this application of algebra appears to be indeted, yet fun as the simplicity of the operations, and the general nature of its theorems, that investiga tions, especially in the higher parts of geometry, are generally order and more expeditions by the alge- bracel method, though lef elegant than by what is purely commented. The comentations albo, and ana- legies of the two fiences established by this application, have given rifle to many curious tpecutions.

Region region-15; provider order 15.

Geometry has been rendered far more extenfire and ufeful, and algebra itself.has received considerable improvements.

Region region-16; provider order 16.

I. Of the Algebra Expertise of Geometrical Mag- minder.

Region region-17; provider order 17.

A line, whether known or unknown, is reprefented by a fragile letter: a _relcangle_ is properly exprefiled by the product of the two letters reprefening its tides: and a _relcangular probability_] by the product of three factors (wo of which represent the fields of any) its rectangular bafes, and the third the altitude.

Region region-18; provider order 18.

There are the molth people copifications of geometrical magnitudes, and any other which has a known proportion to thefe, may in like manner be exprefled at gebranically. Convertively the geometrical magnitudes, reprefected by fich algebraic quantities, may be found, only the global dimensions above the third, one having any corresponding geometrical dimensions, multi be exprefed by proportionals (4).

Region region-19; provider order 19.

The opposite polition of friginal lines, it has been remained, may be exprefiled by the first + and—

Region region-20; provider order 20.

Thus, let a point A be given in the line

Region region-21; provider order 21.

AP, any fragment AP taken to the right hand being considered as politive, a fragment Ap to the left is properly

Region region-22; provider order 22.

(a) All algebrical dimentions above the third multi be exprefiled by inferior geometrical dimensions; and though any algebraic quantities of two or three dimensions may be immediately exprefiled by furfaces and folds respectively, yet it is generally necefary to expref them, and till superior dimensions, by lines.

Region region-23; provider order 23.

If, in any geometrical investigation by algebras, each line is expelled by a finite letter, and each surface or fold by an algebrical quantity of two or three dimensions respectively, then whatever legitimate operations

are performed with regard to them, the terms in any equation derived will, when properly reduced, be all of the fame dimension, and any fuch equation may be cafily exprefilled geometrically by means of proportionals, as in the following example.

Region region-24; provider order 24.

Thus, if the algebraic equation q-f+f=z=-pXI, to to be exprefed geometrically, a, f, o, and d, being fup- pointed to repetis flingful here, let a1 e:cf/p, a contained proposition, then so f'd', c/g and a' c'+f+ b! :a:at+g; then let a:c:i:k;l, and a:c:c:a:l; alfo, let c:dim m:p, and c'd':e:p, or c' c' −d':e:ce-p. By combining the two former proportions (Chap. II. Part I.), c':a'+b':::i:at+g, and combining the latter with this lift found, c'−d':a'+b':e-pXI:cxat+g; therefore c-pXI=cxat+g, and ecc-p:i:i:at+g.

Region region-25; provider order 25.

IF

Region region-26; provider order 26.

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